

Project ID:

J26-IT-334

1. Topic (12 words max)

Integrated Flood and Landslide Risk Assessment and Response System for Sri Lanka

2. Research group the project belongs to

Center of Excellence for AI (CEAI)

3. Specialization of the project belongs to

Artificial intelligence

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Sri Lanka is highly vulnerable to natural disasters, particularly floods and landslides, due to intense monsoon rainfall, rapid urbanization, deforestation, changing land-use patterns, and climate variability. These disasters frequently result in loss of life, damage to infrastructure, disruption of transportation networks, and significant economic losses, especially in districts such as Ratnapura, Kegalle, Badulla, and Nuwara Eliya [1], [2], [3]. According to the Sendai Framework for Disaster Risk Reduction, strengthening disaster preparedness through multi-hazard risk assessment and effective decision-making is essential for reducing disaster impacts [4].

Although government agencies such as the Disaster Management Centre (DMC), the Department of Meteorology, and the National Building Research Organization (NBRO) provide early warnings and hazard information, existing systems mainly operate independently and focus on individual hazards rather than providing a comprehensive multi-hazard solution [5], [3]. Previous studies have also highlighted that disaster-related data in Sri Lanka are scattered across multiple organizations, limiting data integration and timely decision-making for disaster management.

Furthermore, disaster response in Sri Lanka remains largely reactive, relying on manual coordination and post-disaster assessments rather than real-time risk analysis and resource planning. This reduces the effectiveness of emergency response and delays critical decision-making during disaster events.

Therefore, there is a need for an AI-driven integrated disaster management framework that combines flood impact assessment, landslide prediction, intelligent decision support, and emergency response coordination into a single platform. Such a system can provide dynamic hazard mapping, accurate impact assessment, optimized resource allocation, and evidence-based decision support, thereby improving disaster preparedness, reducing response time, and minimizing the social and economic impacts of floods and landslides in Sri Lanka [4], [6].

References

- [1] National Building Research Organisation (NBRO), "Landslide Induced Resettlement Information System (LIRIS)," Colombo, Sri Lanka. [Online]. Available: <https://liris.nbro.lk/>.
- [2] ReliefWeb, "Sri Lanka: Floods and Landslides," United Nations Office for the Coordination of Humanitarian Affairs (OCHA). [Online]. Available: <https://reliefweb.int/country/lka>.
- [3] Department of Meteorology, Sri Lanka, "Department of Meteorology Official Website." [Online]. Available: <https://www.meteo.gov.lk/>.
- [4] United Nations Office for Disaster Risk Reduction (UNDRR), *Sendai Framework for Disaster Risk Reduction 2015–2030*. Geneva, Switzerland: UNDRR, 2015. [Online]. Available: <https://www.undrr.org/publication/sendai-framework-disaster-risk-reduction-2015-2030>.
- [5] Disaster Management Centre (DMC), Sri Lanka, "River Water Level and Flood Warning System." [Online]. Available: https://www.dmc.gov.lk/index.php?option=com_dmcreports&view=reports&report_type_id=6.
- [6] United Nations Office for Disaster Risk Reduction (UNDRR), *Global Assessment Report on Disaster Risk Reduction (GAR)*. Geneva, Switzerland: UNDRR. [Online]. Available: <https://www.undrr.org/gar>. Accessed: Jun. 28, 2026.

6. Brief description of Existing Research and Systems (300 words max)

Existing research on disaster management has mainly focused on hazard prediction, susceptibility mapping, and early warning generation for floods and landslides. Recent studies in Sri Lanka have applied Geographic Information Systems (GIS), hydrological modelling, remote sensing, and machine learning techniques to improve hazard identification and disaster preparedness.

Flood prediction systems commonly utilize rainfall records, river water levels, digital elevation models, and land-use information to estimate flood extents and severity. Similarly, landslide susceptibility studies rely on rainfall intensity, slope characteristics, geological information, and soil properties to identify vulnerable regions and generate hazard maps.

Several recent studies have demonstrated the effectiveness of machine learning techniques in disaster management applications. Palliyaguru et al. proposed a Random Forest-based approach for predicting landslide susceptibility in the Kalutara District of Sri Lanka [7]. Similarly, Aththanayake et al. introduced an agentic AI framework for supporting human-centered disaster response systems [8].

Despite these advancements, most existing systems focus on individual hazards and provide limited support for integrated multi-hazard assessment, impact estimation, resource optimization, and emergency response coordination. Consequently, there remains a significant gap in developing a unified AI-driven platform capable of integrating flood prediction, landslide prediction, decision support, and emergency response coordination within a single system for Sri Lanka.

References

- [7] S. T. Palliyaguru et al., "Random Forest as a Novel Machine Learning Approach to Predict Landslide Susceptibility in Kalutara District, Sri Lanka," in *Proc. 20th Int. Conf. Advances in ICT for Emerging Regions (ICTer)*, Colombo, Sri Lanka, 2020.
- [8] S. Aththanayake et al., "ResQConnect: Agentic AI for Human-Centered Disaster Response Systems," in *Proc. SLAAI Int. Conf. Artificial Intelligence (SLAAI-ICAI)*, Sri Lanka, 2025.

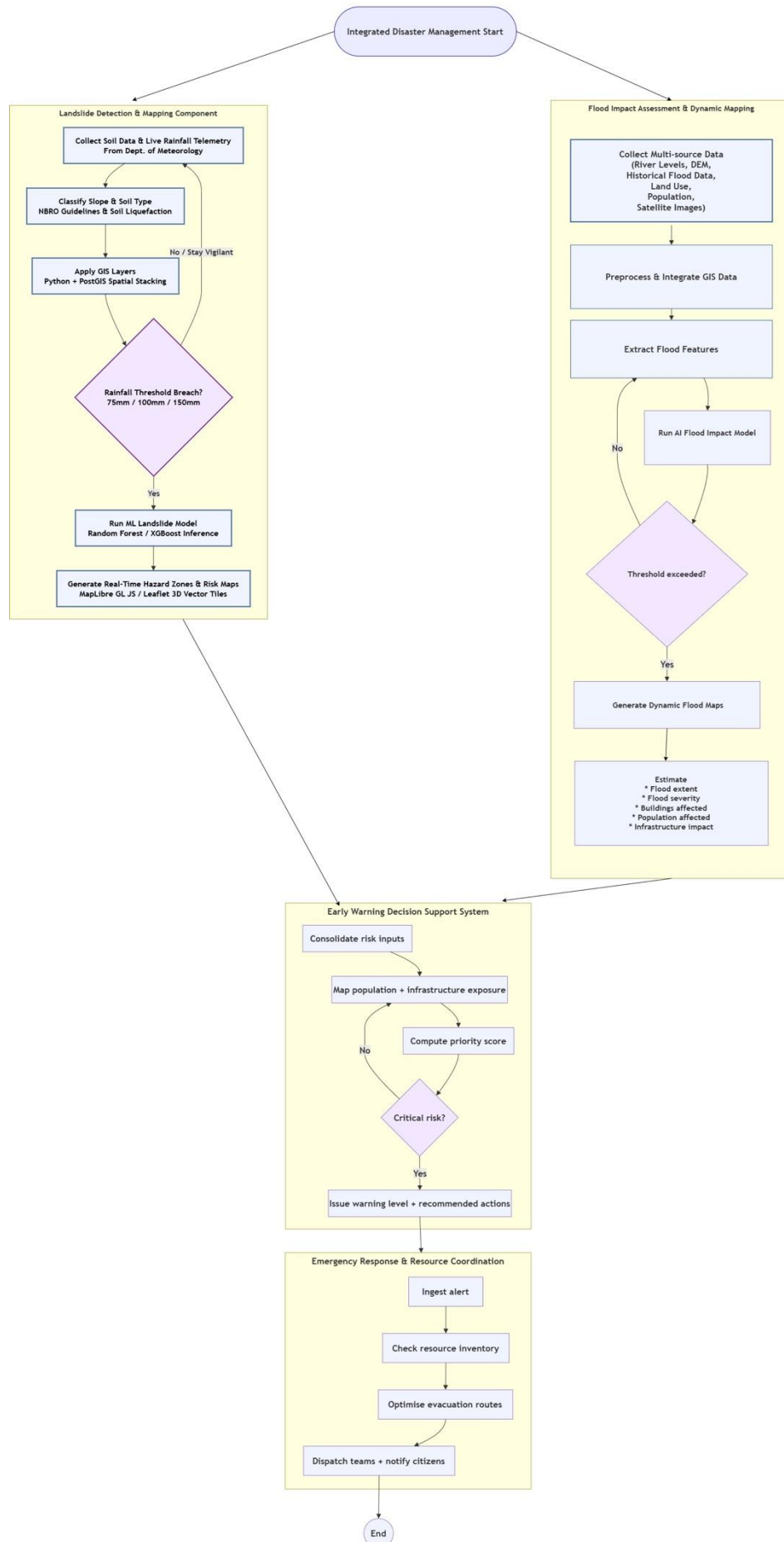
7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

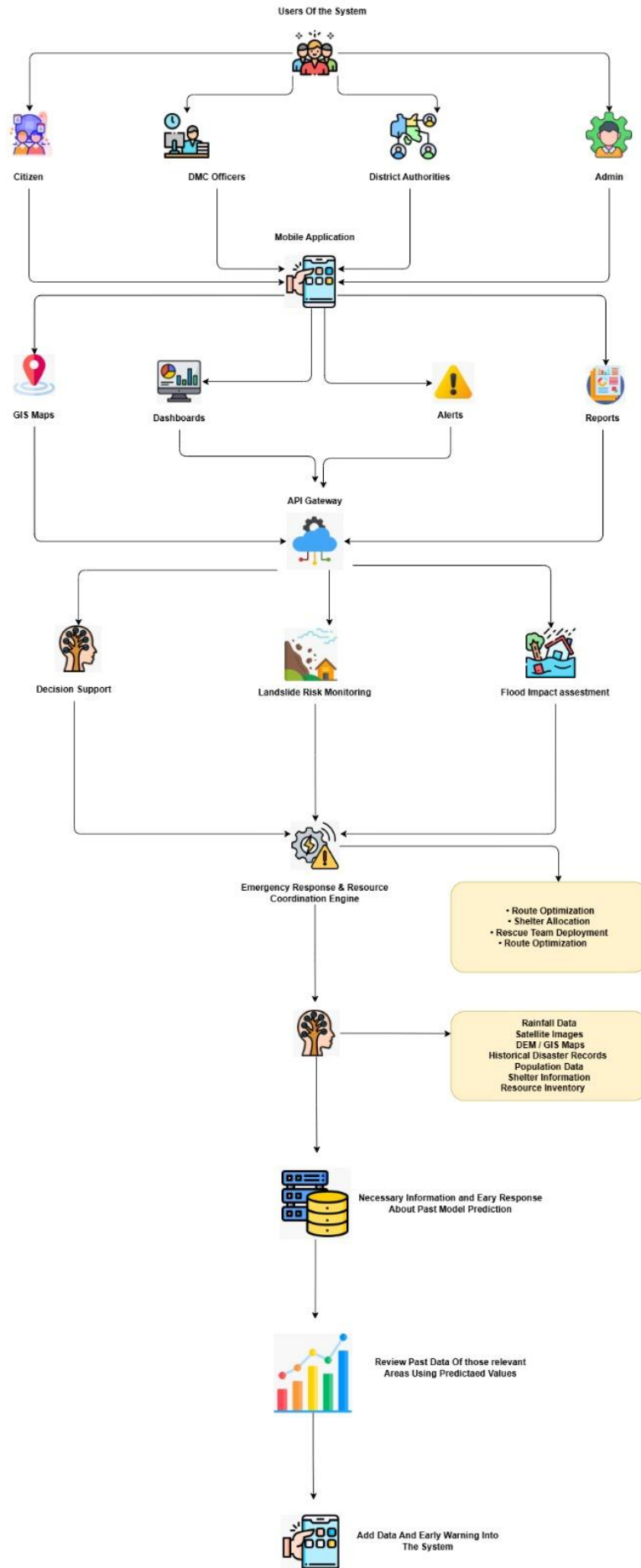
The proposed research project, "Integrated Flood and Landslide Risk Assessment and Response System", requires expertise in Disaster Risk Management, Artificial Intelligence, Geographic Information Systems (GIS), Remote Sensing, Hydrology, and Software Engineering. Team members can gain the necessary domain expertise through an extensive literature review of disaster management research, consultation with academic supervisors, and studying technical reports published by organizations such as the Disaster Management Centre (DMC), National Building Research Organization (NBRO), Department of Meteorology, and Survey Department of Sri Lanka. Understanding the causes, impacts, and management strategies of floods and landslides will help the team design solutions relevant to Sri Lankan conditions.

The technical knowledge and skills required for the project can be acquired through university coursework, online learning platforms, workshops, research papers, and practical experimentation. Team members will strengthen their knowledge in machine learning, deep learning, GIS analysis, remote sensing, database management, web application development, and data visualization. Technologies such as Python, Scikit-learn, TensorFlow, QGIS, PostGIS, Google Earth Engine, and modern web development frameworks will be used throughout the project. The team will also conduct prototype development and testing to gain practical experience with these technologies.

The project requires multiple datasets related to floods and landslides. Historical rainfall records, weather observations, and climate data will be collected from the Department of Meteorology and publicly available sources. Flood records, damage reports, and disaster statistics will be obtained from the Disaster Management Centre (DMC). Landslide hazard maps, rainfall thresholds, slope information, and historical landslide records will be collected from the National Building Research Organization (NBRO). Additional datasets such as Digital Elevation Models (DEM), satellite imagery, land-use data, river water levels, transportation networks, and population statistics will be acquired from sources such as Sentinel, NASA, USGS, OpenStreetMap, and the Survey Department of Sri Lanka. These datasets will be integrated and processed to support flood assessment, landslide prediction, decision support, and emergency response coordination.

The project mainly uses environmental, geospatial, and publicly available disaster datasets; therefore, significant ethical concerns are limited. However, if community surveys, interviews, GPS-based field observations, or citizen feedback are collected during system validation, ethical clearance may be required from the university's ethics review committee. Any personal or sensitive information will be anonymized and handled according to established research ethics and data privacy guidelines.





8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The proposed research, “Integrated Flood and Landslide Risk Assessment and Response System”, requires expertise in Disaster Management, Artificial Intelligence (AI), Machine Learning (ML), Geographic Information Systems (GIS), Remote Sensing, Hydrology, and Software Engineering. Team members can gain the required domain knowledge through literature reviews, consultations with academic supervisors, published research articles, technical documentation, and guidance from organizations such as the Disaster Management Centre (DMC), National Building Research Organization (NBRO), Department of Meteorology, and Survey Department of Sri Lanka. The project focuses on flood impact assessment, landslide prediction, decision support, and emergency response coordination.

The technical knowledge required for the project will be acquired through academic coursework, online learning platforms, workshops, and practical implementation. Team members will develop skills in machine learning, geospatial analysis, remote sensing, database management, API integration, and web application development. Technologies such as Python, TensorFlow, Scikit-learn, QGIS, PostGIS, Google Earth Engine, Leaflet, and MapLibre will be used throughout the project.

The project requires both historical and real-time environmental datasets. Data will be collected from government agencies and publicly available repositories. Key data sources include:

1. [The first version of nation-wide open 3D soil database for Sri Lanka](#)
2. [Sri Lanka - Historical Flood Mapping](#)
3. [Sri-Lanka-Disaster-Data-Resource-Hub](#)
4. [Spatial Data Services](#)
5. [Increasing trend of flood and landslide disasters in Sri Lanka: Socio-Economic perspective](#)
6. [Historical rainfall data for Sri Lanka](#)

These datasets will provide rainfall records, river conditions, elevation models, land-use information, population exposure data, satellite imagery, historical flood events, and landslide hazard information required for AI-based risk assessment and emergency response planning.

The project mainly utilizes environmental, geospatial, and publicly available datasets; therefore, significant ethical concerns are minimal. However, if field surveys, community interviews, GPS observations, or citizen feedback are collected during system validation, ethical clearance may be required from the university's Ethics Review Committee. Any personal information collected will be anonymized and used solely for academic research purposes in compliance with data privacy and research ethics guidelines.

9. Objectives and Novelty

Main Objective To create an intelligent disaster management system that identifies flood and landslide risks, estimates their impacts, and provides timely decision support and emergency response recommendations for Sri Lanka.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Sinhalagoda S.A.U.R.A.V IT23280892	Develop an AI-driven Flood Risk Prediction and Dynamic Mapping framework for Sri Lanka to assess flood severity, estimate potential damage, and generate flood risk maps using geospatial, hydrological, and satellite data.	- Collect historical flood records, river water level data, DEM, land use, population, and satellite imagery. - Preprocess and integrate multi-source datasets. - Develop AI/ML models for flood severity and impact assessment. - Generate flood risk and flood extent maps using GIS techniques. - Analyze affected population, buildings, and infrastructure. - Validate flood predictions using satellite observations. - Implement Explainable AI techniques to identify key flood risk factors. - Develop visualization dashboards and dynamic mapping outputs.	Multi-source Data Fusion: Combines river levels, DEM, land use, population exposure, historical flood records, and satellite imagery in a single AI framework. Impact-focused Assessment: Predicts flood severity and expected damage instead of only predicting flood occurrence. AI + Satellite Validation: Uses real-time satellite imagery to validate and update flood extent maps. Explainable AI: Provides reasons behind flood risk predictions, improving transparency and decision-making.

Bagya R M S IT23394124	To develop a Machine Learning-driven system for early detection and probabilistic prediction of landslide occurrences across highly vulnerable regions in Sri Lanka before disaster strikes.	• Data Acquisition: Formally request and compile historical landslide point locations, hazard zones, and high-resolution precipitation records from national departments. • Geospatial Feature Stacking: Layer digital elevation models for slope angles, bedrock classes, and specific local soil maps using a spatial database backend. • Rainfall Trigger Thresholding: Program categorical machine learning inputs based on official national landslide alert levels and precipitation volumes. • Human Factor Integration: Integrate supplementary risk parameters representing human errors, including dangerous slope cutting, deforestation, and unplanned drainage blocks. • Granular Spatial Triage: Train machine learning classification models to process multi-factor data and map hazard risks from provincial scales down to exact towns and local villages. • High-Performance Rendering: Build a WebGL-powered 3D front-end to visualize dynamic terrain stability indicators smoothly.	Multi-Factor Probability Calculation: Establishing a highly accurate, real-time probability engine that calculates landslide risks by mathematically correlating dynamic rainfall volumes with static slope geometry and soil saturation variables simultaneously. Inclusive and Localized Live Mapping: Designing a Web-GIS live map featuring high-contrast accessible user interface themes for elderly citizens, alongside dual-language localization in native languages for village-level alerts. Explainable Micro-Level Terrain Failures: Transitioning from broad regional warnings to site-specific evaluations by integrating an Explainable AI layer to show the exact percentage breakdown of the environmental and human causes behind a predicted slope failure.
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Prathibhani Y.L.O.L IT23317512	To Develop a disaster risk assessment and early warning decision support system to identify high-priority risk areas and support emergency response planning.	• Collect flood/landslide outputs, DEM, rainfall, census, infrastructure, and historical events; secure data-sharing with providers. • Standardize spatial/temporal references, clean data, generate grid and administrative aggregates. • Combine flood and landslide probabilities into joint hazard scores, model temporal overlap and cascade amplification, propagate uncertainty. • Compute population exposure (density, elderly, children, disabled) and socio-economic vulnerability indices. • Map hospitals, shelters, schools, roads; compute accessibility and facility capacity metrics. • Design priority scoring (weighted scoring/AHP/fuzzy logic or ML ranking) integrating hazard, exposure, vulnerability, and infrastructure. • Translate scores to graded alerts with predefined operational actions and message templates. • Recommend rescue-team/shelter placement and routes using travel-time estimates and predicted road disruptions.	Decision-centered integration: Moves beyond prediction to produce prioritized, actionable decisions for authorities and citizens. Multi-hazard cascading fusion: Formally models flood–landslide interactions and temporal overlap to compute joint risk. Vulnerability- and infrastructure-aware prioritization: Weights human vulnerability and critical facilities to focus life-saving actions. Explainable, confidence-aware alerts: Provides transparent rationales and uncertainty to reduce false alarms and build trust. Operational readiness: Directly maps outputs to resource-placement and routing advice, shortening decision latency in real crises.
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Palihena M.B IT23411562	To develop a real-time emergency response and resource coordination system that converts flood and landslide risk alerts into prioritized dispatch and evacuation decisions for Sri Lanka's disaster management authorities.	• Collect risk alerts from flood, landslide, and climate prediction components via a standardized API. • Consolidate overlapping alerts per district into a single unified severity score. • Map population density onto the risk layer • Identify and overlay vulnerable group data to assess exposure levels. • Query real-time availability of rescue boats, ambulances, shelters, and supply stocks at District Disaster Management Centers. (DDMCs) • Rank affected zones by severity score. • Allocate available resources to zones based on priority ranking. • Generate optimized evacuation routes for citizens. • Generate optimized routes for rescue teams. • Dispatch alerts and shelter information to DMC coordinators via SMS/app. • Dispatch alerts and shelter information to affected citizens via SMS/app. • Track response outcomes post-event. • Use outcome data to improve future resource prioritization decisions.	Automated Resource Dispatch: Directly links disaster predictions to real-time resource allocation and deployment decisions. Integrated Route Optimization: Generates optimized evacuation and rescue routes within a unified disaster management platform. Real-time Response Coordination: Replaces manual, phone-based coordination with an automated and centralized response framework. Prediction-to-Action Integration: Bridges the gap between hazard prediction and emergency response within a single system. End-to-End Disaster Response: Supports the complete response pipeline from risk alert → resource allocation → route dispatch → citizen notification.
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10. This section is to be completed by the Supervisor and the Co-supervisor of the Project.

- a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	
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- b) The proposed topic exhibits novelty.

Yes	✓	No	
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- c) The student group can successfully execute the proposed project.

Yes	✓	No	
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- d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	
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- e) Data required for the project is available.

Yes	✓	No	
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- f) Any other comments:

The proposed project is novel and suitable for a final-year research project. The modular architecture enables each component to be developed, validated, and evaluated independently using historical or simulated datasets, while supporting systematic integration into a unified disaster management platform.

	Title	First Name	Last Name	Signature
Supervisor*	Dr.	Kapila	Dissanayaka	
Co-Supervisor*	Mr.	Uchira	Wickramaratne	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

*- At this stage, identifying the Co-Supervisor is compulsory. Students should reach out to a staff member, with guidance from their supervisor, to serve as the co-supervisor for this project.